8 Week 7: Methods for Adjustment — Matching, Weighting, and IPW

8.1 Design Versus Analysis in Observational Studies

Researchers often emphasize a separation between:

- **Design:** use X and Z (but not Y) to construct a comparison that mimics a randomized experiment (e.g., matching, weighting, trimming).
- **Analysis:** once the design is fixed, analyze outcomes Y using simple estimators (e.g., differences in means, regression).

This separation:

- reduces model dependence,
- clarifies assumptions,
- aligns observational studies with the logic of randomized experiments.

8.2 Matching on Covariates and Propensity Scores

Matching constructs a subset of the data in which treated and control units are comparable on observed covariates.

Matching on X. We can match treated and control units directly on X using:

- exact matching (for discrete covariates),
- Mahalanobis distance matching,
- optimal matching (minimizing total distance).

Matching on $e(X)$. Alternatively, we can match on the estimated propensity score:

- nearest-neighbor matching (with or without replacement),
- caliper matching (restricting matches to be within a tolerance in $e(X)$),
- full matching (forming strata with varying numbers of treated and control units).

Subclassification. Divide units into K subclasses based on quantiles of $e(X)$ (e.g., quintiles). Within each subclass, treated and control units are compared. As $K \to \infty$, subclassification approximates full conditioning on $e(X)$.

8.3 Estimation After Matching

Suppose we have constructed a matched sample (e.g., 1:1 pairs). A simple estimator of the ATE is the difference in means in the matched sample:

$$\hat{\tau}_{\text{match}} = \frac{1}{n_M} \sum_{i:Z_i=1} \big(Y_i - Y_{j(i)}\big),$$

where $j(i)$ is the matched control for treated unit i, and n_M is the number of matched pairs.

Under SUTVA, ignorability, and good covariate balance, $\hat{\tau}_{\text{match}}$ is approximately unbiased for τ.

8.4 Inverse Probability Weighting (IPW): Setup and Assumptions

We now turn to weighting methods. IPW reweights units to create a pseudo-population in which treatment is independent of covariates.

We observe i.i.d. units

$$\{(Y_i, Z_i, X_i)\}_{i=1}^N,$$

with $Z_i \in \{0, 1\}$ and pre-treatment covariates X_i. Potential outcomes $(Y_i(1), Y_i(0))$ are defined as before.

We impose:

- **Consistency:** $Y_i = Z_i Y_i(1) + (1 - Z_i) Y_i(0)$.
- **Ignorability:** $(Y_i(1), Y_i(0)) \perp Z_i \mid X_i$.
- **Overlap:** $0 < e(X_i) = \Pr(Z_i = 1 \mid X_i) < 1$.

Our target estimand is the ATE:

$$\tau = \mathbb{E}[Y(1) - Y(0)].$$

8.5 The IPW Estimator and Its Intuition

Define the IPW weights

$$w_i = \frac{Z_i}{e(X_i)} + \frac{1 - Z_i}{1 - e(X_i)}.$$

The IPW estimator of the ATE is $1/N \sum_i w_i Y_i$ which we can write in its more familiar form

$$\hat{\tau}_{\text{IPW}} = \frac{1}{N} \sum_{i=1}^N \left(\frac{Z_i Y_i}{e(X_i)} - \frac{(1 - Z_i) Y_i}{1 - e(X_i)} \right).$$

Intuition. Each unit is weighted by the inverse probability of receiving the treatment it actually received. Units that were unlikely to receive their observed treatment (e.g., treated units with low $e(X)$) are upweighted to represent similar units who did not receive that treatment. This is analogous to the Horvitz–Thompson estimator in survey sampling.

8.6 Unbiasedness of IPW (with Known Propensity Score)

Assume the true propensity score $e(X)$ is known. We show that $\hat{\tau}_{\text{IPW}}$ is unbiased for τ.

Consider the treated component:

$$\mathbb{E}\left[\frac{ZY}{e(X)}\right] = \mathbb{E}\left[\mathbb{E}\left(\frac{ZY}{e(X)} \mid X\right)\right].$$

By consistency ($Y = Y(1)$ when $Z = 1$),

$$\mathbb{E}\left(\frac{ZY}{e(X)} \mid X\right) = \mathbb{E}\left(\frac{ZY(1)}{e(X)} \mid X\right).$$

By ignorability, Z and $Y(1)$ are conditionally independent given X, so

$$\mathbb{E}\left(\frac{ZY(1)}{e(X)} \mid X\right) = \frac{1}{e(X)} \mathbb{E}[Z \mid X] \mathbb{E}[Y(1) \mid X].$$

But $\mathbb{E}[Z \mid X] = e(X)$, so

$$\mathbb{E}\left(\frac{ZY(1)}{e(X)} \mid X\right) = \mathbb{E}[Y(1) \mid X].$$

Taking expectations over X,

$$\mathbb{E}\left[\frac{ZY}{e(X)}\right] = \mathbb{E}[Y(1)].$$

Analogously,

$$\mathbb{E}\left[\frac{(1-Z)Y}{1-e(X)}\right] = \mathbb{E}[Y(0)].$$

Therefore,

$$\mathbb{E}[\hat{\tau}_{\text{IPW}}] = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = \tau.$$

8.7 Consistency and Asymptotic Normality

Under standard regularity conditions and overlap, the IPW estimator is a sample average of i.i.d. terms with finite variance. By the law of large numbers,

$$\hat{\tau}_{\text{IPW}} \xrightarrow{p} \tau.$$

By the central limit theorem,

$$\sqrt{N}(\hat{\tau}_{\text{IPW}} - \tau) \xrightarrow{d} \mathcal{N}(0, V_{\text{IPW}}),$$

where V_{IPW} is the variance of the influence function

$$v_i = \frac{Z_i Y_i(1)}{e(X_i)} - \frac{(1-Z_i)Y_i(0)}{1-e(X_i)} - \tau.$$

8.8 Variance and the Problem of Extreme Weights

The variance of $\hat{\tau}_{\text{IPW}}$ depends on:

- the variability of $Y(1)$ and $Y(0)$,
- the distribution of $e(X)$.

When $e(X_i)$ is close to 0 or 1, the weights $1/e(X_i)$ or $1/(1-e(X_i))$ become large, leading to:

- high variance,
- sensitivity to misspecification of $e(X)$,
- poor finite-sample behavior.

This instability motivates stabilized weights and overlap-focused designs (e.g., trimming units with extreme propensity scores).

8.9 Covariate Balance as a Theorem

Define group-specific weights:

$$w_i^{(1)} = \frac{Z_i}{e(X_i)}, \qquad w_i^{(0)} = \frac{1-Z_i}{1-e(X_i)}.$$

Proposition 3. *For any integrable function $g(X)$,*

$$\mathbb{E}[w^{(1)} g(X)] = \mathbb{E}[g(X)], \qquad \mathbb{E}[w^{(0)} g(X)] = \mathbb{E}[g(X)].$$

Proof. Consider the treated weights:

$$\mathbb{E}[w^{(1)}g(X)] = \mathbb{E}\left[\frac{Zg(X)}{e(X)}\right] = \mathbb{E}\left[\mathbb{E}\left(\frac{Zg(X)}{e(X)} \mid X\right)\right].$$

Conditioning on X,

$$\mathbb{E}\left(\frac{Zg(X)}{e(X)} \mid X\right) = \frac{g(X)}{e(X)}\mathbb{E}[Z \mid X] = \frac{g(X)}{e(X)}e(X) = g(X).$$

Taking expectations over X,

$$\mathbb{E}[w^{(1)}g(X)] = \mathbb{E}[g(X)].$$

The control case is analogous.

□

Interpretation. IPW constructs two weighted groups whose covariate distributions both match the full population distribution. This is the precise sense in which IPW achieves covariate balance.

8.10 Stabilized Weights

To reduce variance, we define stabilized weights:

$$\tilde{w}_i^{(1)} = \frac{Z_i \Pr(Z=1)}{e(X_i)}, \qquad \tilde{w}_i^{(0)} = \frac{(1-Z_i)\Pr(Z=0)}{1-e(X_i)}.$$

These weights preserve the balancing property while shrinking extreme values. In practice, stabilized weights often yield:

- smaller variance,
- more stable finite-sample performance,
- more interpretable effective sample sizes.

8.11 Augmented IPW (AIPW) and Double Robustness

AIPW combines outcome modeling with IPW. Let $\hat{m}_1(X)$ and $\hat{m}_0(X)$ be estimates of $\mathbb{E}[Y \mid Z = 1, X]$ and $\mathbb{E}[Y \mid Z = 0, X]$, respectively. The AIPW estimator is

$$\hat{\tau}_{\text{AIPW}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{Z_i(Y_i - \hat{m}_1(X_i))}{e(X_i)} - \frac{(1-Z_i)(Y_i - \hat{m}_0(X_i))}{1-e(X_i)} + \hat{m}_1(X_i) - \hat{m}_0(X_i)\right].$$

Doubly robust property. Under SUTVA, ignorability, and overlap, $\hat{\tau}_{\text{AIPW}}$ is consistent for τ if either:

- the propensity score model $e(X)$ is correctly specified, or
- the outcome regression models $\hat{m}_1(X)$ and $\hat{m}_0(X)$ are correctly specified.

This "double robustness" provides protection against misspecification of one component.

8.12 Diagnostics and Practical Considerations

Key diagnostics for matching and weighting designs include:

- standardized mean differences (before and after adjustment),
- variance ratios,
- overlap in estimated propensity scores,
- distribution of weights (e.g., histograms, effective sample size),
- sensitivity to trimming or truncation of extreme weights.

Design-based thinking emphasizes:

- constructing a design that achieves good balance,
- diagnosing balance quantitatively,
- only then analyzing outcomes.

8.13 Conceptual Takeaways

- SUTVA ensures that potential outcomes are well-defined.
- Ignorability replaces randomization with an assumption about covariates.
- Overlap is essential for identification and stable estimation.
- Propensity scores balance observed covariates in a low-dimensional space.
- Matching and weighting are design tools, not mere modeling tricks.
- IPW reweights the sample to mimic a randomized experiment.
- AIPW provides doubly robust estimation by combining models.
- Causal inference without randomization requires explicit assumptions, careful design, and transparent diagnostics.